

Ridwan Babatunde Ayinde

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| [Certifications](#) | [My Portfolio](#)

PROFESSIONAL SUMMARY

Data engineer with a quantitative foundation in Mechanical Engineering, currently building applied pipeline and data infrastructure skills through an MSc in Operations Research at the University of Lagos and the MIT Emerging Talent Program. Hands-on experience constructing geospatial data pipelines, processing large-scale operational datasets with Python, and applying clustering algorithms to structure raw data for downstream use. Comfortable working across the full data workflow, from acquisition and cleaning through transformation and delivery. Seeking data engineering and analytics engineering roles where strong fundamentals and a drive to build scalable, reliable data systems are valued.

TECHNICAL SKILLS

Languages & Processing	Data & Infrastructure	Tools & Methods
• Python (Pandas, NumPy) • Data Pipeline Construction • Geospatial Data Processing • Graph / Network Data • ETL Fundamentals	• SQL (querying & joins) • Data Cleaning & Wrangling • Relational Databases • Large Dataset Handling • Data Quality & Validation	• Git & Version Control • OpenStreetMap / OSM • Scikit-learn (ML basics) • Matplotlib (output viz) • Excel (data prep), Linux (CLI)

KEY PROJECTS

Last-Mile Logistics Data Pipeline - Lagos, Nigeria [\[View Project\]](#)

- Designed and executed an end-to-end data pipeline ingesting geospatial inputs, service boundaries, road networks, and delivery point coordinates, from raw OpenStreetMap exports.
- Built a routable road network graph in Python to model real Lagos delivery constraints, transforming unstructured OSM data into a queryable graph structure.
- Applied K-means clustering as a transformation step to partition delivery points into logical route zones, reducing downstream processing load.
- Delivered clean, structured outputs ready for route optimisation algorithms and visual reporting, demonstrating a full extract-transform-load (ETL) workflow.

U.S. Domestic Flights Data Processing & Analysis [\[View Project\]](#)

- Ingested and processed a large-scale flight operations dataset using Python and Pandas, handling missing values, type conversion, and multi-field joins across airline, airport, and temporal dimensions.
- Engineered delay pattern features by airline, route, and seasonal period to enable structured comparative analysis across the dataset.
- Produced visualisation outputs in Matplotlib to surface trends for operational review, bridging raw data processing and business-readable output.

Airline Customer Segmentation Pipeline [\[View Project\]](#)

- Cleaned and structured transactional customer data, resolving inconsistencies and standardising travel behaviour fields for clustering input.
- Applied K-means clustering using Scikit-learn to segment customers, tuning cluster count based on within-cluster variance analysis.

- Exported structured segment outputs with profiled attributes to support demand planning and resource allocation decisions.

PROFESSIONAL EXPERIENCE

Data & Cost Analyst (Engineering Projects)

Aug 2023 – Present

CA Consultant LTD — Lagos, Nigeria

- Structured and maintained project cost datasets across multiple HVAC installations, ensuring data consistency and traceability under a Quality Management System (QMS).
- Compiled and cross-referenced Bills of Quantities (BOQ) requiring systematic data extraction, transformation, and numerical validation across supplier pricing sources.
- Built comparative pricing models from vendor data to support procurement decisions — an applied form of structured data aggregation and evaluation.
- Managed documentation pipelines ensuring all project records were complete, audit-ready, and accessible — developing habits of data quality and governance.

EDUCATION

MSc, Operations Research (Part-Time)

Jan 2026 – Present

University of Lagos (UNILAG) — Lagos, Nigeria

Focus: mathematical optimisation, quantitative modelling, and algorithmic approaches to operational systems, directly applicable to data engineering problem-solving.

MIT Emerging Talent Program — Computer & Data Science

Nov 2024 – Dec 2025

Massachusetts Institute of Technology (MIT) — Online

Covered Python programming, computational thinking, data structures, and applied data science methods.

B.Eng, Mechanical Engineering

Oct 2021

Federal University of Technology, Akure (FUTA) — Nigeria